SMART ENERGY METERING

AND

THEFT DETECTION SYSTEM

Smart energy metering and theft detection system

Engineering Clinics Fall Semester 2025-26: Review-2

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CONTRIBUTION

01. DHANA LAKSHMI:
- Leader
- Research
- Software
- Content for ppt

02. DHEERAJ:
- Hardware
- Online purchase of components
- Research
- Content for ppt

03. CHANAKYA:
- Hardware
- Offline purchase of components
- Research
- Designing ppt

04. BHAVYA:
- Software
- Offline purchase of components
- Research
- Designing ppt

05. HIRANMAI:
- Documentation
- Picture Collection for ppt
- Offline Purchase of Components
- Expenditure tracking

06. SRI LAKSHMI:
- Documentation
- Research
- Expenditure Tracking
- Picture Collection for ppt

COMPONENTS

ESP-32
Cost- 479/-

Current Sensor
Cost- 130/-

Voltage Sensor
Cost- 160/-

Relay Module
Cost- 45/-

Hall Effect Sensor
Cost- 80/-

LCD Display
Cost- 320/-

I2C
Cost- 150/-

ESP-32

- Brain of the System
- low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth

WORKING

- Reads data from the sensors
- Sends data wirelessly to the monitoring app or cloud
- Triggers alerts or disconnects supply if theft is detected.

Current Sensor

Voltage Sensor

CURRENT SENSOR

WORKING:

- Detects current
- Sends data through analogue voltage signal to ESP-32

ESP-32

Voltage Sensor

VOLTAGE SENSOR

WORKING:

- Detects voltage
- Sends data through analogue voltage signal to ESP-32

ESP-32

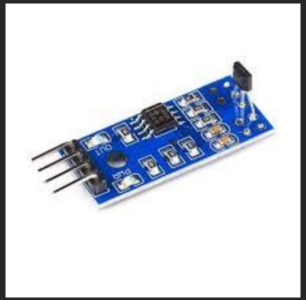
Current Sensor

RELAY MODULE

WORKING:

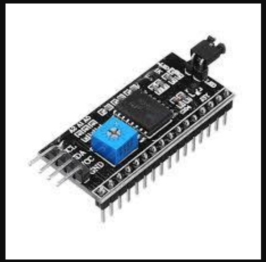
- Controls the AC Mains Supply
- Receives digital analogue signal from ESP-32

Hall Effect Sensor



LCD Display

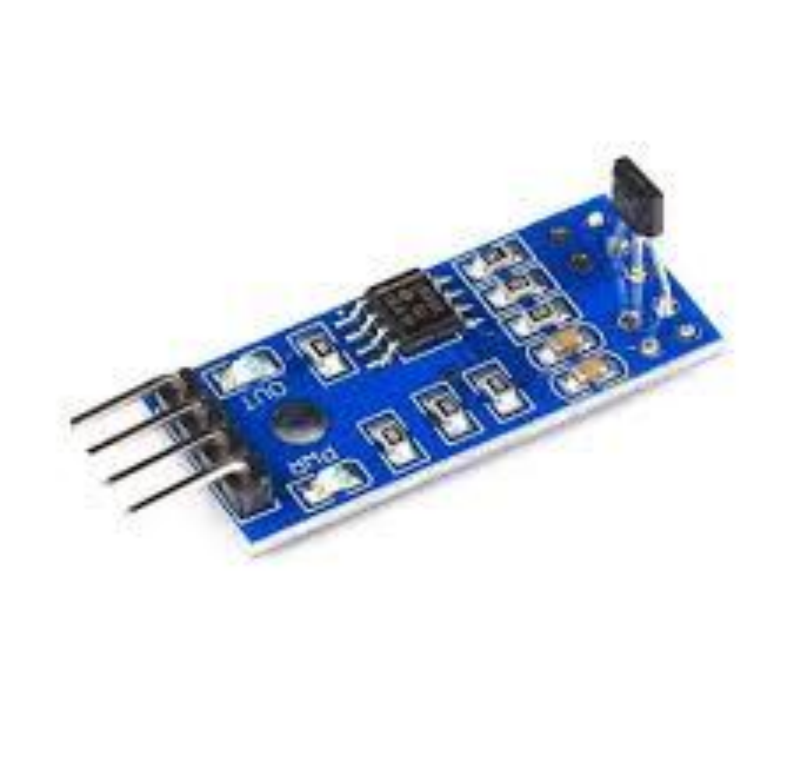
I2C



HALL EFFECT SENSOR

WORKING:

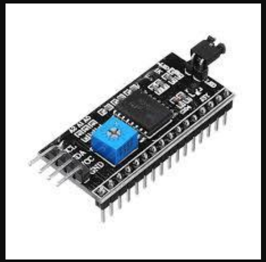
- Detects magnetic field
- Sends data through analogue voltage signal to ESP-32



Relay Module

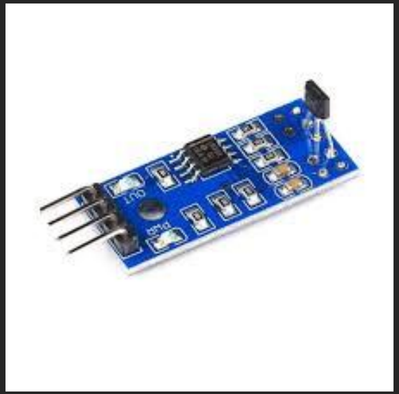
LCD Display

I2C



Relay Module

Hall Effect Sensor

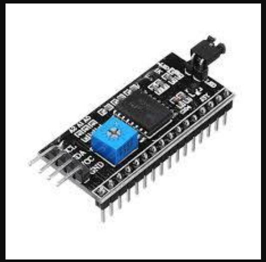


LCD DISPLAY

WORKING:

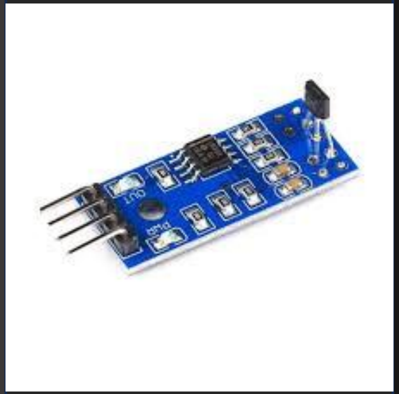
- Displays output characters

I2C



Relay Module

Hall Effect Sensor

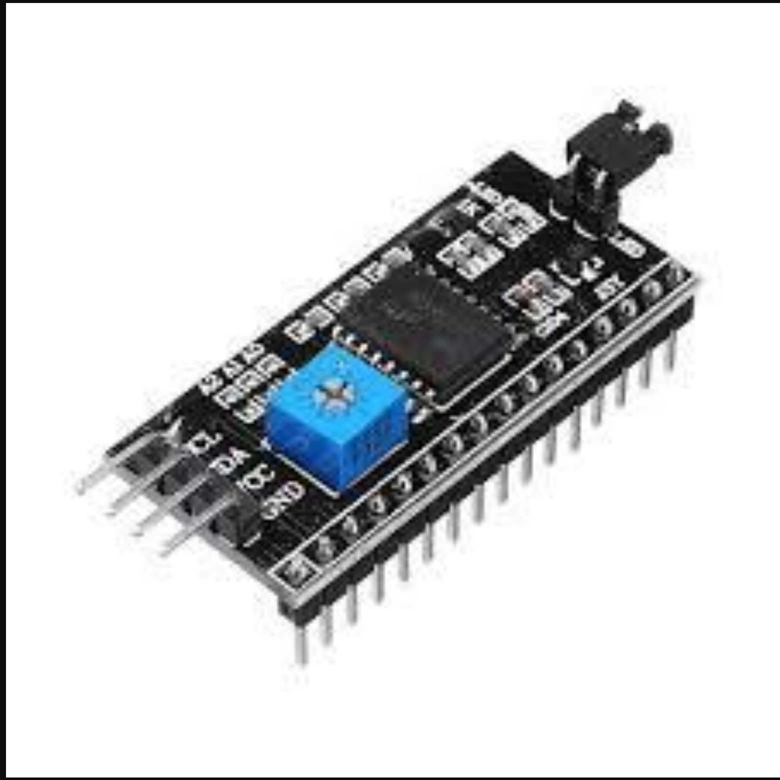


LCD Display

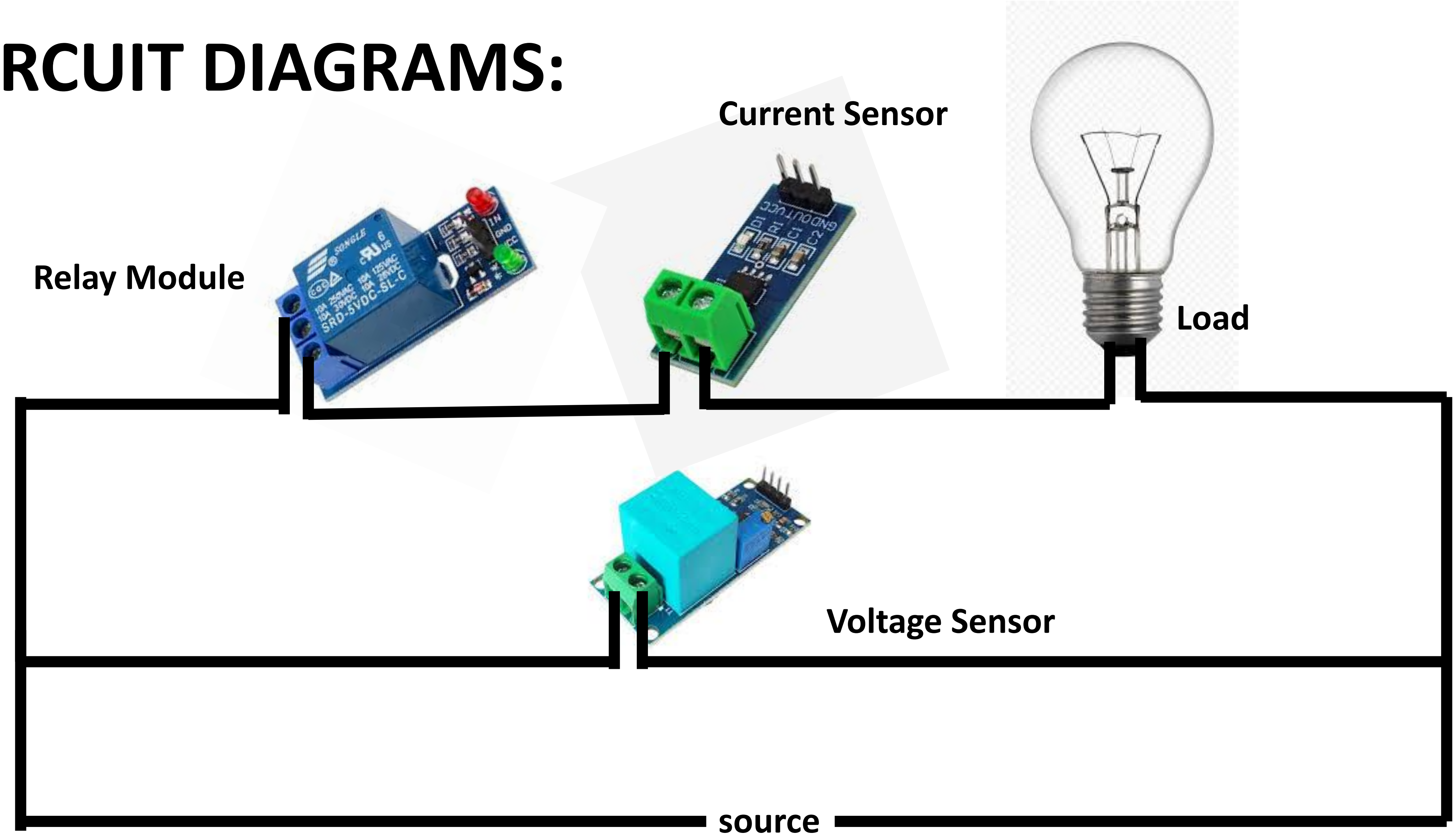
I2C (Inter-Integrated Circuit)

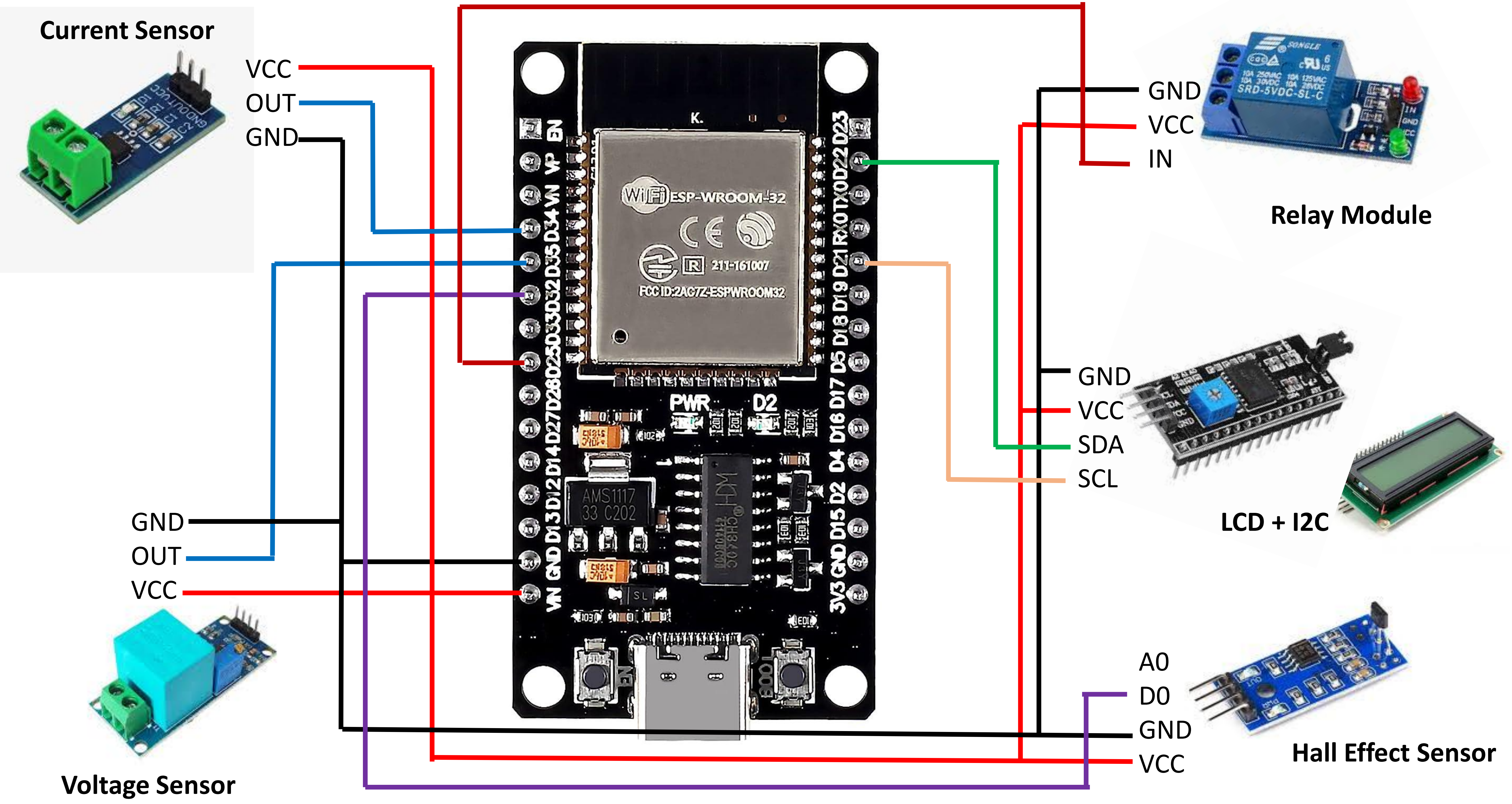
WORKING:

- Communicates between ESP-32 and LCD



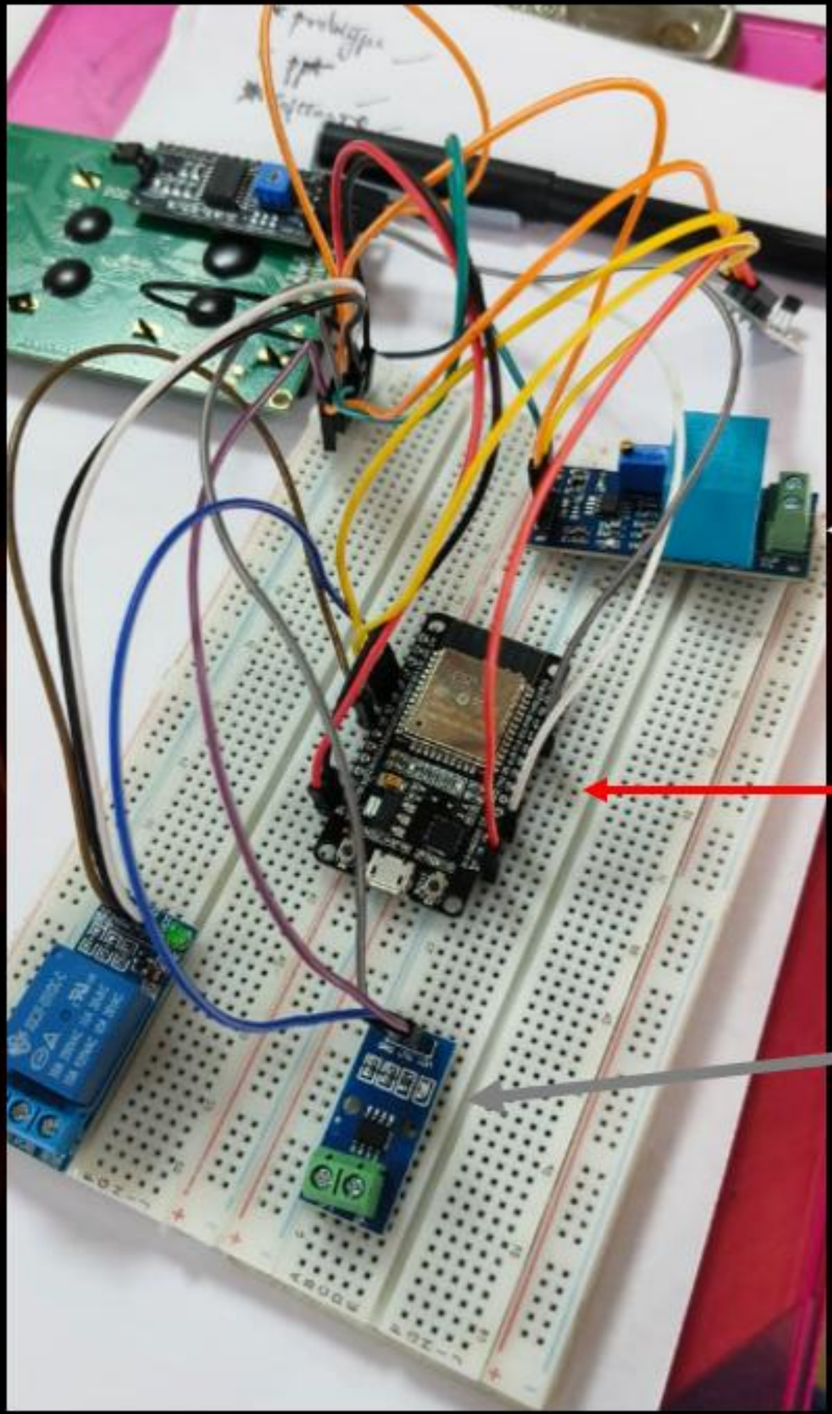
CIRCUIT DIAGRAMS:





Current Sensor
VCC
OUT
GND
GND
OUT
VCC
Voltage Sensor
GND
VCC
IN
Relay Module
GND
VCC
SDA
SCL
LCD + I2C
A0
D0
GND
VCC
Hall Effect Sensor

Prototype



LCD+I2C

CONNECTIONS:
- SDA -> D21
- SCL -> D22
- GND -> GND
- VCC -> Vin

Voltage Sensor

CONNECTIONS:
- GND -> GND
- OUT -> D35
- VCC -> Vin

HALL EFFECT SENSOR

CONNECTIONS:
- D0 -> D32
- GND -> GND
- VCC -> Vin

RELAY MODULE

CONNECTIONS:
- IN -> D25
- GND -> GND
- VCC -> Vin

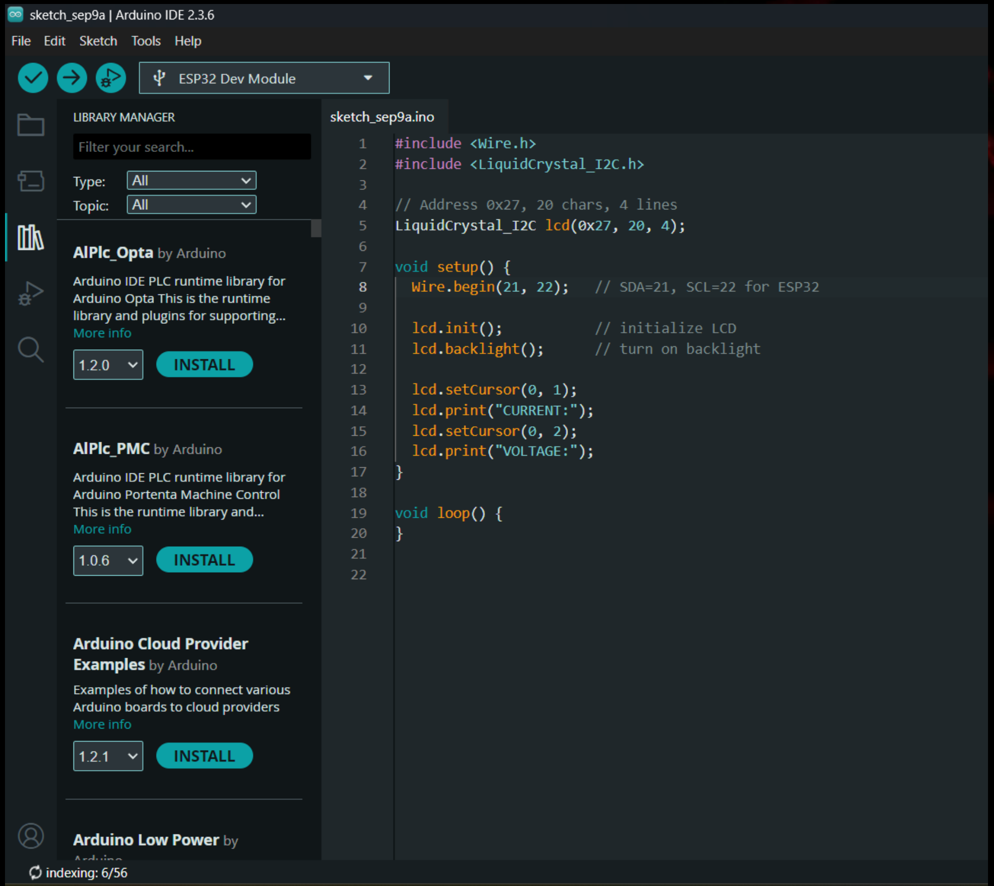
ESP-32

Current Sensor

Connections
- VCC -> Vin
- OUT-> D34
- GND-> GND

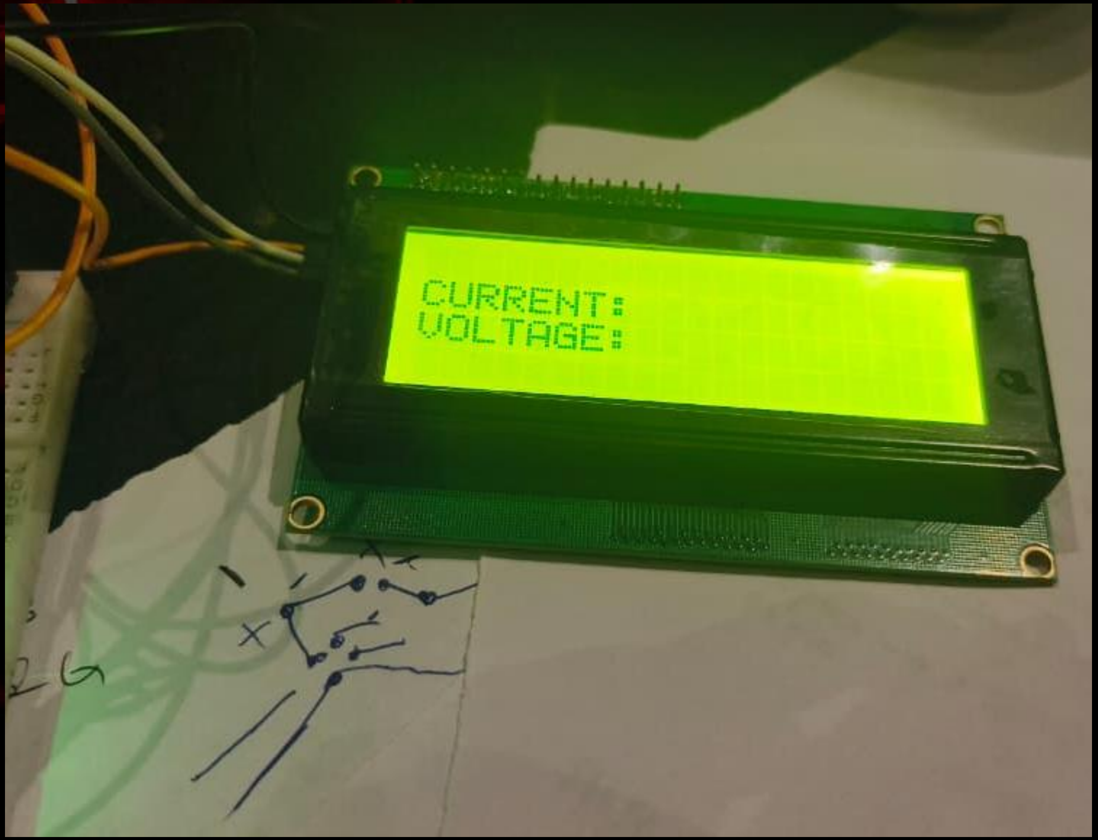
SOFTWARE

- IDE used: Arduino IDE
- Board type : ESP32 Dev Module
- Simulator : Proteus, wokwi.com



TEST CODE

OUTPUT



TIMELINE OF PROJECT (Review-1 to Review-2)

1ST WEEK

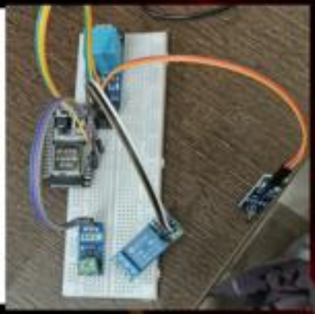
- No. of Meetings: 2 (12th August and 15th August)
- Discussion on purchase of components

2nd WEEK

- No. of Meetings: 0 (CAT-1 was going on)
- Research on the project
- Online purchase
- Offline purchase

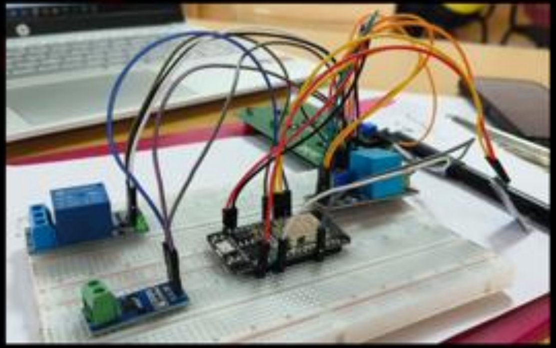
3rd WEEK

- No. of Meetings: 1 (28th August)
- Components examination



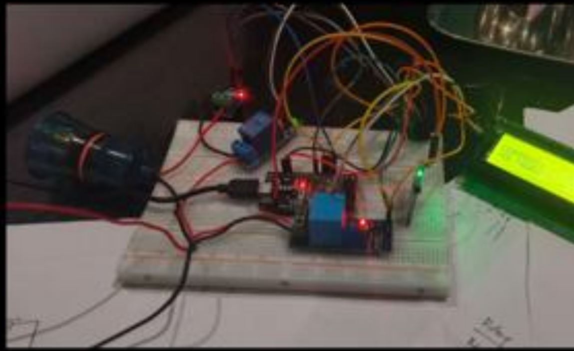
4th WEEK

- No. of Meetings: 2 (2nd and 4th September)
- Hardware Updates and explanation



5th WEEK

- No. of Meetings: 2 (9th and 10th September)
- 9TH SEPTEMBER: Software Discussion
- 10TH SEPTEMBER: Final Discussion



Further Enhancements:

- Software (completion of coding)
- Software integration using Blynk
- Placing the final circuit in an enclosure

THANK YOU!